

Task 15 — Time-Series Forecasting

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Advanced · Advanced

Level 15 of 20 · ramp 71%

1 · What this is & why it matters

Build time-series forecasts with seasonality and validation to support planning over the coming periods.

Reliable forward-looking numbers drive staffing, capacity and budget decisions across the business.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A validated seasonal forecast with intervals and documented assumptions.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Decomposition

separating trend, seasonality and residual.

Seasonality

repeating patterns (weekly, monthly, yearly).

Stationarity

stable statistical properties over time, needed by some models.

Backtesting

validating on rolling historical windows.

Forecast intervals

communicating uncertainty as a range.

5 · Recommended stack & tools

- statsmodels (SARIMA) / Prophet
- pandas
- matplotlib

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Decompose the series into trend/seasonal/residual.
2. Choose a model that fits the observed structure.

3. Backtest on rolling windows, comparing to a naive baseline.
4. Tune for the best validated error.
5. Produce forecasts with intervals.
6. Document drivers, assumptions and risk periods.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- SARIMA vs Prophet vs ML regressors with lag features.
- Single global model vs per-segment models.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- Which seasonal pattern matters most for planning?
- Where will the forecast most likely be wrong, and what's the cost?
- Is added model complexity actually earning lower error?

9 · Definition of Done

Done when all are true and demoable

- A validated seasonal forecast with intervals and documented assumptions.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A validated seasonal forecast with intervals and documented assumptions.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Ignoring seasonality.
- No backtest, just an in-sample fit.
- Hiding uncertainty.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design
- Time-series and segmentation techniques